Finn Beruldsen

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Personal Statement

Ph.D. candidate in Biochemistry with a focus on computational biology. I seek to leverage computational tools in biophysics and immunoinformatics to enhance the safety, efficacy, and accessibility of cell-based immunotherapies. I develop multi-scale models and software to improve our understanding of how molecular interactions involving T-cell receptors manifest and drive patient outcomes.

Education

University of Houston
Ph.D. in Biochemistry — 2022 – Present
Advised by Dinler Antunes, Presidential Fellow
GPA: 4.0/4.0

University of Iowa
Visiting Student — 2021
Distinction in Biostatistics
GPA: 4.0/4.0

Hendrix College
Bachelor of Arts in Biochemistry & Molecular Biology — 2017 – 2021
Phi Beta Kappa, Magna cum laude, Distinction in Biochemistry and Molecular Biology
GPA: 3.9/4.0

Research Experience

University of Houston, Center for Nuclear Receptors and Cell Signaling
Supervisor: Dinler A. Antunes
Project 1: The molecular basis of the T-cell cross-reactive response in hepatitis E

- Performed RMSD calculations, binding affinity predictions, and heuristic generation to evaluate T-cell cross-reactivity.
- Presented a novel framework for predicting cross-reactivity based on biochemical and structural properties.
- Developing gene-to-sequence-to-structure pipeline using webserver automation.

Project 2: Developing methodology for performing steered molecular dynamics to incorporate biophysical metrics in T-cell activation prediction

- Prototyped SMD scripts with TCRpMHC structures.
- Defined additional factors for metrics of force resiliency.

- Tested dynamic network analysis with protein structures.

Project 3: Developing novel strategies and visualizations for Molecular Dynamics

- Created Time Series RMSF, known as RMSX, published in Scientific Reports
- Combined the features of RMSF and RMSD to see when and where changes occur in a simulation as a plugin for the MDAnalysis package.

Hannover Medical School (Medizinische Hochschule Hannover), Twincore
Supervisors: Anke Kraft PhD, Markus Cornberg MD
Project: Immunological Methods in Chronic and Acute Viral Diseases

- Conducted T-cell cultures, flow cytometry, and DNA isolation.
- Contributed to research on lymphocytic choriomeningitis virus (LCMV) genotypes.

University of Iowa, College of Public Health
Supervisor: Patrick Breheny PhD
Project: Sleep Apnea and Bone Marrow Transplant Recovery

- Tracked longitudinal patient data and created statistical models for sleep apnea and immune outcomes following bone marrow transplants.

Hendrix College
Supervisor: Dr. Heidi Dahlmann
Project: Carbonyl Olefin Metathesis Catalyst Synthesis

- Synthesized catalysts and characterized them using chromatography and NMR.
- Presented findings at ACS Sci-Mix and Arkansas State Capitol.

University of Arkansas for Medical Sciences
Supervisors: Andres Caro PhD, Lawrence Cornett PhD
Project: Translational research for neonatal opioid withdrawal syndrome (NOWS)

- Analyzed maternal-infant opioid metabolite levels using tandem MS/MS.
- Assisted in developing policies for the NICU to treat opioid-exposed infants.

Colorado State University Veterinary Teaching Hospital
Supervisors: Kirk McGilvray PhD, Howard B. Seim III, DVM, DACVS
Project: Quantifying osteogenesis in bone implants

- Prepared samples for biomechanical testing and generated 3D bone density maps.

Rice University
Supervisor: Zachary Ball
Project: Organorhodium complexes: synthesis and characterization

- Performed mass spectrometry high-performance liquid chromatography on synthesized organometallic complexes.

Honors and Awards

- James P. Taylor Foundation for Open Science Scholarship, full conference fee and airfare support, Galaxy Community Conference (GCC2026), Clermont-Ferrand, France (2026)
- ITCR Training Network Trainee Travel Award, NCI Informatics Technology for Cancer Research (ITCR) Annual Meeting, Cold Spring Harbor Laboratory, NY (2026)

- NSM Graduate Student Conference Travel Award, College of Natural Sciences and Mathematics, University of Houston (2025, 2026)
- Department of Biology and Biochemistry Student Travel Award, University of Houston (2025, 2026)
- NIH/NCI Travel Award (2025)
- Distinguished Graduate Student Leadership Award (2025)
- University of Houston Presidential Fellow (2022)
- University of Iowa ISIB Fellow (2021)
- American Chemical Society Organic Chemistry Research Award (2021)
- Phi Beta Kappa (2021)
- Hendrix Student Impact Award (2021)
- Distinction in Biochemistry & Molecular Biology (2021)
- American Chemical Society, Division of Organic Chemistry Undergraduate Fellow (2020)

Technical Skills

- **Bioinformatics Tools**: RMSD calculations, binding affinity predictions (PRODIGY), supervised and unsupervised machine learning, clustering, sequence-based analysis comparison, application of ordinary differential equations (ODEs) to molecular systems and cellular systems
- **Programming Languages**: Python, R, Bash, Mathematica, C++ (Arduino)
- **Laboratory Techniques**: NMR, Mass-spectrometry, Chromatography (TLC, HPLC, Column), Organic Synthesis, DNA Processing, Flow Cytometry/FACS, Cell Culture
- **Molecular Modeling**: Steered Molecular Dynamics (NAMD, GROMACS, MDAnalysis), Molecular Visualizations (VMD, Pymol, ChimeraX, Blender), Molecular Modeling (AlphaFold, TCRmodels)

Leadership & Service

- President - Bioscience Graduate Society, University of Houston (2023-Present)
- Co-Chair, Department of Biology and Biochemistry Graduate Awards Committee (2024–2025)
- Director, Moderator - 14th-17th Annual Bioscience Graduate Society Research Symposium (2025)
- Senate Chair of Mental Health, Hendrix College

Publications & Presentations

1. S. Klein, J. Mischke, **F. Beruldsen**, I. Prinz, D.A. Antunes, M. Cornberg, A.R.M. Kraft. (May 2023). *T Cell Immune Responses Are Shaped Differently during Chronic Viral Infection.* Pathogens, 12(5).

2. **F. Beruldsen**, Vaz de Freitas M., D.A. Antunes. *Structure-guided Unsupervised Classification of T-cell Cross-reactivity Patterns: Alternative Recognition vs Molecular Mimicry.* Manuscript in preparation for *Bioinformatics* (Oxford).

3. **F. Beruldsen**, Vaz de Freitas M., D.A. Antunes. *Applications of Ordinary Differential Equations to T-cell Efficiency and Competition.* Manuscript in preparation for *Frontiers in Immunology* (Hypotheses and Theory).

4. Vaz de Freitas, **F. Beruldsen**, M., Le H.N., D.A. Antunes. *Exploring Cross-reactivity as a Potential Therapeutic Strategy for Chronic Hepatitis E Virus (HEV) Infection.* Manuscript in preparation for *Pathogens* (MDPI).

5. **F. Beruldsen**, C. Soon, A.F. Fonseca, M. Cornberg, D.A. Antunes. (December 12, 2022). *Analysis of T-cell Receptor Interactions Reveals Alternative Recognition of HEV-derived Peptides.* CNRCS Annual Symposium, University of Houston, Houston, TX.

6. **F. Beruldsen**, H.N. Le, C. Soon, A.F. Fonseca, M. Cornberg, D.A. Antunes. (October 21, 2022). *Alternative Recognition Patterns Driving Cross-genotype Response with Hepatitis E Virus-specific T Cells.* 32nd Keck Annual Research Conference Structural Biology: Past, Present, and Future, Rice University, Houston, TX.

7. **F. Beruldsen**, A.F. Fonseca, D.A. Antunes. (October 21, 2022). *Enabling Large-Scale Modeling of Immune Complexes with Webserver Automation Using Selenium.* 32nd Keck Annual Research Conference Structural Biology: Past, Present, and Future, Rice University, Houston, TX.

8. **F. Beruldsen**, C. Soon, A.F. Fonseca, M. Cornberg, D.A. Antunes. (December 7, 2021). *Explaining Directionality in T-cell Cross-reactivity between HEV Genotypes.* Center for Nuclear Receptors and Cell Signaling Virtual Symposium, University of Houston, Virtual.

9. **F. Beruldsen**, L. Brinna, P. Breheny. (July 22, 2021). *Sleep Apnea and Bone Marrow Transplant Recovery.* University of Iowa Summer Research Virtual Symposium, Virtual.

10. **F. Beruldsen**, L. Mortan, H. Dahlmann. (February 19, 2020). *Organocatalyzed Carbonyl-Olefin Metathesis: Catalyst Reactivity and Substrate Scope.* Research Posters at the Arkansas State Capitol, Arkansas State Capitol Building, Little Rock, AR.

11. **F. Beruldsen**, H. Dahlmann. (September 22, 2020). *Optimization of Organocatalyzed Carbonyl-Olefin Metathesis Reactions.* American Chemical Society Division of Organic Chemistry Virtual SURF Presentations, Virtual.

12. H.A. Dahlmann, P. Crook, **F. Beruldsen**. (August 22, 2022). *Organocatalyzed Carbonyl-Olefin Metathesis: Exploring Catalyst Reactivity and Substrate Scope.* 264th Annual ACS National Meeting, McCormick Place, Chicago, IL. (Also selected to be presented as a Sci-Mix poster).

13. **F. Beruldsen**, M. Vaz de Freitas, D.A. Antunes. (April 13, 2024). *Combining the Best Features of RMSD and RMSF: RMSX, Showing When and Where Changes Happen in Molecular Dynamic Simulations.* The 28th Sealy Center for Structural Biology and Molecular Biophysics Annual Symposium, University of Texas Medical Branch, Houston, TX. (A manuscript is also in revision)

14. **F. Beruldsen**, E. R. Espinosa (February, 25th 2025). *3D Printing and Computer-Aided Design - An Introduction for Researchers in Life Sciences.* Society for the Advancement of Chicanos/Hispanics and Native Americans in Science Seminar, Houston, TX.

15. **F. Beruldsen**, (February, 28th 2025). *Visualizing and Quantifying Molecular Changes Over Time - an Introduction to QwikMD, RMSD, RMSF and RMSX.* Graduate Structural Bioinformatics (6397), Guest Lecturer University of Houston, Houston, TX.

16. **F. Beruldsen**, D.A. Antunes. (August 8, 2025). *RMSX and Flipbook: Tools to See When and Where Proteins Move During MD Simulations.* Flash talk at the 63rd Hands-On Workshop

on Computational Biophysics, Auburn University, Auburn, AL.

17. **F. Beruldsen**, M. Vaz de Freitas, D.A. Antunes. (August 21, 2025). *Flipbook & RMSX: Publication Ready Molecular Visualizations.* Oral presentation at the NCI Informatics Technology for Cancer Research (ITCR) Annual Meeting, Rockville, MD.

18. **F. Beruldsen**, D.A. Antunes. (November 4, 2025). *Mathematical modeling of interclonal T-cell competition reveals impact of suboptimal clones on tumor clearance.* Journal for ImmunoTherapy of Cancer, 13(Suppl 2):A1232. Abstract 1082, SITC 40th Annual Meeting. Poster presentation, National Harbor, MD. https://doi.org/10.1136/jitc-2025-SITC2025.1082

19. **F. Beruldsen**, M. Vaz de Freitas, D.A. Antunes. (February 20, 2026). *High resolution mapping of protein motions in time and space with RMSX and Flipbook.* Scientific Reports. https://doi.org/10.1038/s41598-026-39869-7

20. P. Borges, M. Vaz de Freitas, J. Yoo, **F. Beruldsen**, et al. (June 2026). *Mapping the TCR landscape: computational tools empowering translational immunology and therapy design.* Journal for ImmunoTherapy of Cancer, 14(6):e014184. https://doi.org/10.1136/jitc-2025-014184

21. **F. Beruldsen**. (2026). *Existing Multi-Material FFF Printers Are Sufficient For (Near) Arbitrary Coloration.* Preprint, SSRN. Under review at *Computers & Graphics*; Ms. Ref. No. CAG-D-26-00574. https://doi.org/10.2139/ssrn.6876564

22. H.A. Dahlmann, **F. Beruldsen**, H.L. Criswell, P.F. Crook, E.C. Glassford, A.K. Jones, R.B. Mitchell, L.F. Mortan, N. Ryan, A.M. Smith, and E.M. Vazquez. (2026). *Phosphoric Acid Derivative-Catalyzed Carbonyl-Olefin Metathesis. Organics.* Accepted / in press.

23. **F. Beruldsen**, D.A. Antunes. (June 2026). *When and Where Proteins Move: RMSX and Flipbook for Molecular Dynamics.* Short oral presentation and poster, Galaxy Community Conference (GCC2026), Clermont-Ferrand, France.

24. **F. Beruldsen**, D.A. Antunes. (July 15–17, 2026). *Composition-Aware Modeling of T-Cell Clonal Competition in Cancer Immunotherapy.* Poster presentation, NCI Informatics Technology for Cancer Research (ITCR) Annual Meeting, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY.